

Fig. 2: Actual-size PCB layout of dual-channel active AC analogue probe

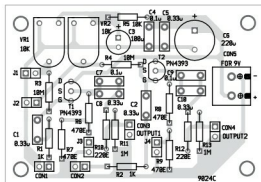


Fig. 3: Components layout for the PCB

jumpers J1 and J2, and connect resistors R3 and R4 to ground.

The gain of each of the channels is slightly below unity, but that is not important in most of the cases. Most oscilloscopes produce a square wave reference signal, which can be used for measurement of the gain and evaluation of the probe.

Construction and testing

An actual-size PCB layout for dual-channel active AC analogue probe is shown in Fig. 2 and its components layout in Fig. 3. After assembling the circuit on the PCB, enclose it in a suitable cabinet with connectors CON1 and CON2 affixed on the front side and connectors CON3 and CON4 on the rear side of the cabinet.

The probe can be implemented on a small PCB. During testing and adjustment, the inputs of the probe can be connected or soldered to L-C networks using very short (3-4cm long) conductors.

Power the circuit using a dry battery or rechargeable battery. The circuit does not require any special adjustment, so you can start using it straightaway after assembly. If you use jumpers J3 and J4, adjust VR1 and VR2 for the best performance of the probe.

After proper implementation, the probe can have bandwidths above 10MHz with a signal generator having low output impedance. **EFY**



Petre Tzy Petrov was a researcher and assistant professor in Technical University of Sofia (Bulgaria) and expert-lecturer in OFFPT(Casablanca), Kingdom of Morocco. Now he is working as an electronics engineer in the private sector in Bulgaria

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